Simulation and Generative Models for Robust Wearable ECG: Physics, Multiphysics, PINNs and Diffusion, Measured on Real Sensor Noise

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Abstract

Wearable physiological sensing is limited by scarce labels and by motion artifacts that clinical training data do not contain. We study whether simulation and generative modeling can scale training beyond the available real data, and we measure the answer on real data with real recorded sensor noise. Using single-lead PTB-XL ECG and MIT-BIH NSTDB noise under a leakage-free protocol, we build (i) a differentiable ECG dynamical model calibrated to 3,907 real beats (median R^2=0.95); (ii) an electro-mechanical electrode–skin multiphysics model (contact mechanics + electro-quasi-static finite volume, verified to 2×10^{-4} against an analytic solution); (iii) a physics-based sensor/artifact model; (iv) an inverse PINN; (v) a conditional 1-D diffusion model; and (vi) a physics-informed neural surrogate of a finite-element electrode solver, used for gradient-based electrode design. Sensor-model augmentation raises AUROC under 0 dB real noise from 0.654 to 0.818, exceeding augmentation with real recorded noise (0.807). We diagnose a sampler failure that saturated 40% of diffusion samples and trace it to the terminal signal-to-noise ratio of the cosine schedule. The fix needs no retraining: saturation drops to 0.5%, real-vs-synthetic discriminator accuracy falls from 0.96 to 0.63, and a classifier trained only on generated ECG reaches 0.829 AUROC (real-trained: 0.801). Generated data combined with sensor augmentation gives the best result: 0.838 clean and 0.827 AUROC at 0 dB. The surrogate reaches 0.02% median impedance error with calibrated uncertainty and is 37,714× faster than the solver. Ten percent of the real data plus 8,000 generated records matches training on all of it (AUROC 0.822 vs. 0.824 clean; 0.803 vs. 0.802 at 0 dB). We also report where methods fail: the PINN beats filtering only at low SNR, multiphysics-derived sensor priors did not improve the downstream model, and limit-cycle physics guidance of the diffusion sampler hurt fidelity.

1 Problem and protocol

Task. NORM vs. abnormal (MI/STTC/CD/HYP) from one 10-s lead (lead II, 100 Hz, a wearable-like view of PTB-XL [1]); 5,137 records with the official folds: 1–8 train (3,960), 9 validation, 10 test (592). **Noise.** MIT-BIH NSTDB [2] electrode-motion, muscle and baseline-wander recordings. Each recording is split in time: the first half is the only noise any training run may use; the second half only corrupts the test set. **Low-data regime.** A stratified 10% of the training records; the simulator bank, diffusion models and classifier in this regime see only these records. All downstream numbers are mean±std over 3 seeds, with model selection on the clean validation fold.

2 Methods

Differentiable physics simulator. The McSharry ECG model [3] drives a cardiac phase $\theta(t)$ with an RR process (Mayer + respiratory components) and integrates $\dot{z} = -\sum_i a_i \Delta\theta_i e^{-\Delta\theta_i^2/2b_i^2} - (z - z_0)$. Because the z-equation is linear, we integrate it *exactly* for zero-order-hold forcing with an FFT convolution (checked against RK4). This makes the simulator batched and differentiable. We calibrate the 16 parameters of every real median beat in parallel by gradient descent through the simulator, with *variable projection*: amplitudes and offset enter linearly and are solved by ridge least squares, which removes a local minimum in which inverted T-waves collapse instead of flipping (Fig. 1).

Multiphysics electrode model. Contact mechanics (asperity contact for dry electrodes, elastic gel and stratum-corneum compression, partial lift-off) sets a thin-layer interface admittance, coupled to a cell-centred finite-volume solve of $\nabla \cdot (\sigma^* \nabla \phi) = 0$ with complex conductivity in layered tissue (Fig. 2). Verification: analytic thin-strip error 2×10^{-4}, reciprocity 10^{-10}, grid-convergence spread 7×10^{-5}.

Sensor model. Poisson road/body events drive a damped seat/torso response $d(t)$, which produces an AC-coupled skin-stretch potential, impedance-modulated gain $Z_{in}/(Z_{in} + Z_e(t))$, impedance-mismatch mains pickup (60 Hz aliased to 40 Hz), EMG bursts, electrode lift-off to the rail and ADC effects. It is FFT-based and batched, so it runs inside the training loop.

Inverse PINN. A Fourier-feature MLP $s(t)$, a low-order baseline $w(t)$ and learnable ODE parameters minimise $\|y - s - w\|^2 + \lambda\|\dot{s} - F(\theta;\phi) + s\|^2$, with $\dot{s}$ by auto-

Table 1: Downstream AUROC under real held-out sensor noise (3 seeds). †original diffusion sampler.

Training data (100% real)	clean	6 dB	0 dB
real only	.812±.009	.762±.010	.654±.009
+ real NSTDB noise (ref.)	.823±.009	.820±.008	.807±.010
+ **sensor model**	.833±.004	.829±.004	.818±.007
+ simulator data	.830±.008	.808±.015	.743±.021
+ diffusion data†	.835±.005	.814±.014	.756±.019
+ diffusion + sensor†	.836±.003	.831±.002	.827±.007
+ **diffusion + sensor (fixed)**	**.838**±.001	**.834**±.003	**.827**±.003
10% real			
real only	.834±.002	.794±.011	.705±.030
+ sensor model	.809±.003	.803±.006	.790±.007
+ sim-pretrained diff. + sensor	.806±.008	.804±.007	.787±.005

Table 2: Generative fidelity vs. held-out real test data. FED: Fréchet distance on classifier embeddings; Disc.: real-vs-synthetic discriminator accuracy (0.5 ideal); TSTR: train on synthetic, test on real.

Source	FED↓	LSD dB↓	Disc.↓	TSTR↑
real (train split)	4.4	0.20	0.50	0.802
calibrated simulator	6.6	8.80	0.95	0.808
original sampler				
diffusion, full real	72.9	5.44	0.96	0.819
diffusion, 10% real	107.3	1.80	0.93	0.812
diffusion, sim-pre. + 10%	43.2	3.19	0.86	0.808
fixed sampler (leading timesteps, selected on validation)				
diffusion, full real	1.3	0.34	0.63	0.829
diffusion, sim-pre. + 10%	2.0	2.34	0.71	0.825

grad at stochastic collocation points. Segments are solved in parallel with batched (einsum) weights, warm-started from a hard-constraint differentiable-simulation fit.

Conditional diffusion. A 1-D U-Net with FiLM conditioning on diffusion time, diagnosis and heart rate, and self-attention at the bottleneck. Cosine-schedule DDPM [4], EMA, DDIM sampling [5], classifier-free guidance [6]. For the low-data regime we pre-train on 8,000 calibrated-simulator records and fine-tune on the 10% real subset.

3 Results

Robustness (Table 1). The physics-based sensor model is the largest single effect. Under 0 dB real noise it adds +0.164 AUROC with full data, more than augmenting with the real recorded noise itself, and in the 10% regime it adds +0.085. Diffusion data improves clean and moderate-noise accuracy. The combination is best on every test condition. In the 10% regime, augmentation trades ∼0.025 clean AUROC for robustness.

Generative fidelity (Table 2). All synthetic sources carry label information: TSTR AUROC matches or exceeds training on real data (0.801). The calibrated simulator matches the embedding statistics closely (FED 6.6) but not the spectrum (LSD 8.8 dB). With 10% of the data and the original sampler, simulation pre-training cut the diffusion model's FED 2.5× (107→43) and lowered discriminator accuracy (0.93→0.86). Heart-rate conditioning lands within 9.6 bpm of the request with the original sampler and 4.1 bpm with the fixed one. **Sampler diagnosis.** About 40% of the original samples were pinned to the ±12 clip rail. Guidance was the obvious suspect, but the failure persisted at w=0 (39%). The cause is the sampler: DDIM started at t=T−1, where the cosine schedule gives $\bar{\alpha} \approx 2.4 \times 10^{-9}$, so the $\hat{x}_0$ estimate amplifies ϵ-errors $\sim 2 \times 10^4$. Starting at the standard leading timestep ($\bar{\alpha} \approx 9 \times 10^{-4}$) with per-sample dynamic thresholding [8], selected on the validation fold, reduces saturation to 0.5% and validation FED from 49.4 to 0.84, with no retraining (Fig. 3). On test, discriminator accuracy falls from 0.96 to 0.63 and TSTR rises to 0.829. With the fixed sampler, diffusion data plus sensor augmentation reach 0.838±.001 clean and 0.827±.003 at 0 dB. The pre-training comparison above used the original sampler and was not repeated, so it is indicative only.

PINN. On simulated data with known parameters (6 dB sensor-model noise), the PINN recovers P/T-wave strength within 4–9% and timing within 0.03 rad, and raises output SNR from 9.0 to 11.7 dB. QRS parameters are poorly identifiable at 100 Hz (R strength error 35%). On real ECG with real noise, the physics prior wins at 0 dB (7.7 vs. 6.3 dB for band-pass). At 6 and 12 dB band-pass wins (13.9 / 21.0 dB vs. 12.2 / 16.3 dB), because a five-event single-template ODE cannot represent all real morphology.

Multiphysics. The model predicts 21.5 kΩ (wet) vs. 443 kΩ (dry) two-electrode impedance at 10 Hz, near-zero pressure sensitivity for gel but $\kappa \approx 0.4$ for dry electrodes, and lift-off sensitivity $\kappa \approx 0.6$ for both. These fall partly outside the sensor model's hand-set priors. Widening the priors to the multiphysics range, however, did *not* improve the classifier (0.801±.013 vs. 0.818±.007 at 0 dB). The augmentation benefit is robust to the exact priors.

Real in-car ECG. On PhysioNet drivedb [7] (18 real drives, 7,995 windows), with thresholds set to 90% specificity on PTB-XL validation, the real-only model raises false cardiac alarms on 13.9% of in-car windows (PTB-XL NORM test: 12.4%), so it transfers. No augmentation reduced in-car false alarms. The changes were +1.5 to +3.5 points, and every 95% driver-bootstrap interval includes zero. The in-car ECG is clean (beats detectable in 99.9% of windows), and false alarms concentrate in a few drivers (median driver 0.5–1.6%). That points to lead placement and morphology shift, not noise, as the remaining gap.

Neural surrogate (Fig. 4). A P1 finite-element solver (which agrees with the finite-volume solver to 0.46%) generates 4,000 impedance spectra (21 frequencies, 1 Hz–10 kHz) over 10 Latin-hypercube parameters: layer thicknesses and conductivities, stratum-corneum conductivity and permittivity, electrode width, spacing and lift-off. A 5-member deep ensemble with β-NLL variance heads predicts $\log_{10}|Z|$ and phase. The *hybrid* variant learns only the correction to the analytic contact impedance $Z_c = (w_A^{-1} + w_B^{-1})/(L\,Y_{sc})$. On 1,000 held-out spectra its median $|Z|$ error is 0.02% (plain ensemble

0.10%, quadratic ridge 1.67%, prior alone 8.7%), matching the solver's own coarse-vs-fine mesh difference (0.015%). With 500 simulations it beats the plain model trained on 4,000. Its 95% intervals cover 98% of values; with every parameter range widened 25%, the error rises to 0.16% and the predicted σ rises 4.1×. It runs in 8.6 μs per spectrum vs. 325 ms for the solver.

Design. Projected gradient ascent through the surrogate maximises $\mathbb{E}[\ln s - \frac{1}{2}\ln \mathrm{Re}\, Z(10\,\mathrm{Hz})]$ over electrode width w and spacing s ($s{+}w \le 35$ mm, gap ≥ 5 mm) across 512 tissue/lift-off states. Re-scored by the FE solver on 48 independent tissues, the result is within 0.13 dB of a brute-force FE grid optimum; the landscape is flat, and a "maximise spacing" layout loses 1.3 dB.

How much real data does synthesis replace? (Fig. 5) With sensor augmentation in every run and generators trained only on the 10% subset, 10% real plus 8,000 diffusion records reaches 0.822 clean / 0.803 at 0 dB, vs. 0.824 / 0.802 with 100% real data. Smaller generated sets (500–4,000) and physics-simulator records at any size do not close the gap. The real-only curve is flat from 10% to 50% and seeds vary by about ±0.01, so this is a trend, not an exchange rate.

Negative result: limit-cycle guidance. Pushing the sampler's $\hat{x}_0$ toward one-period invariance at the conditioned heart rate improved validation FED (0.84→0.50) at λ=0.05, but on test it worsened FED (1.05→2.64) and LSD (0.37→2.05 dB) for a 0.14 bpm heart-rate gain. Real RR variability makes period averaging blur the QRS.

Second modality: phone IMU (human behaviour). On UCI HAR (waist-worn phone, 6 activities, 9 test subjects never seen in training; raw accelerometer incl. gravity and gyroscope), a physics IMU sensor model (rigid re-orientation of both sensors, calibration errors, loose-mount wobble, clock mismatch) raises macro-F1 from 0.919 to 0.966 on recorded test data and from 0.163 to 0.795 when each test window is re-oriented (an exact physical identity for a rigid phone, though one the augmentation was designed for). Orientation alone accounts for most of it (0.962). A class-conditional IMU diffusion model gives noisy samples (Fréchet distance 115 vs. 2.7 for real data) that nonetheless add accuracy (0.957) but no orientation robustness.

4 Limitations

PTB-XL is resting clinical ECG, and NSTDB noise was recorded on stationary subjects. The multiphysics model is 2-D, uses nominal literature tissue properties and quasi-static contact mechanics, and is written in Python/SciPy (finite volume, with a finite-element check) rather than COMSOL. Results use one dataset and one task.

5 Reproducibility

The script `scripts/run_sensing_gen.sh` downloads the data and reproduces every number (about 4–5 h on an Apple M-series laptop). Results are written to `outputs/sensing_gen/`, and the repository has 28 sensing_gen tests (78 in total) and CI.

References

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[7] J. Healey, R. Picard, Detecting stress during real-world driving tasks using physiological sensors, *IEEE T-ITS*, 2005.

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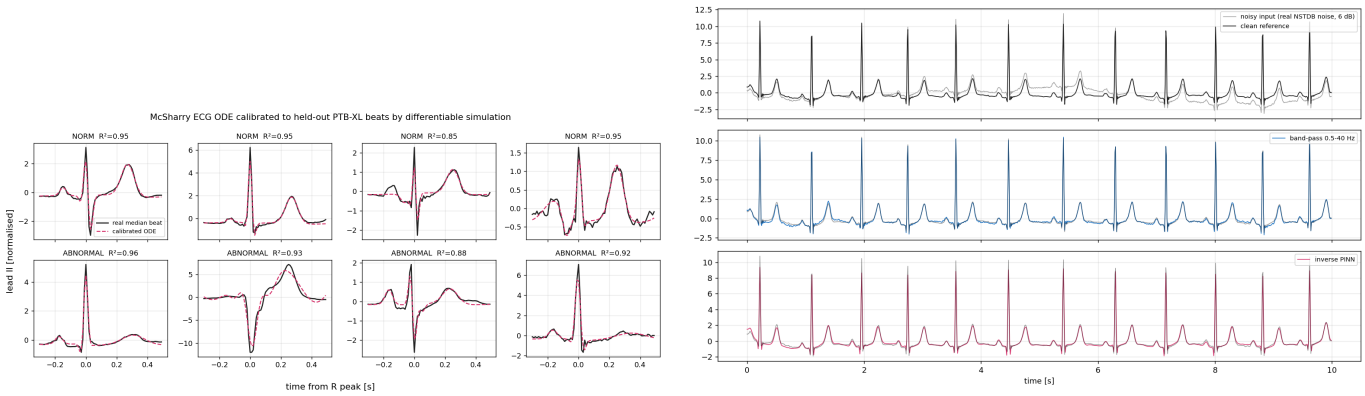


Figure 1: Left: calibrated ODE vs. held-out real median beats. Right: PINN denoising of real ECG under real 6 dB NSTDB noise.

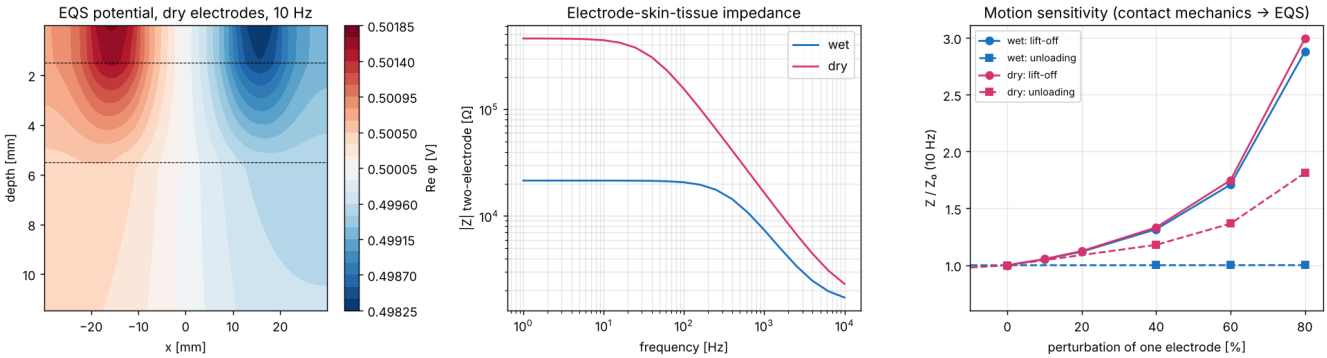


Figure 2: Multiphysics electrode model: EQS potential in layered tissue, impedance spectra, and motion sensitivity.

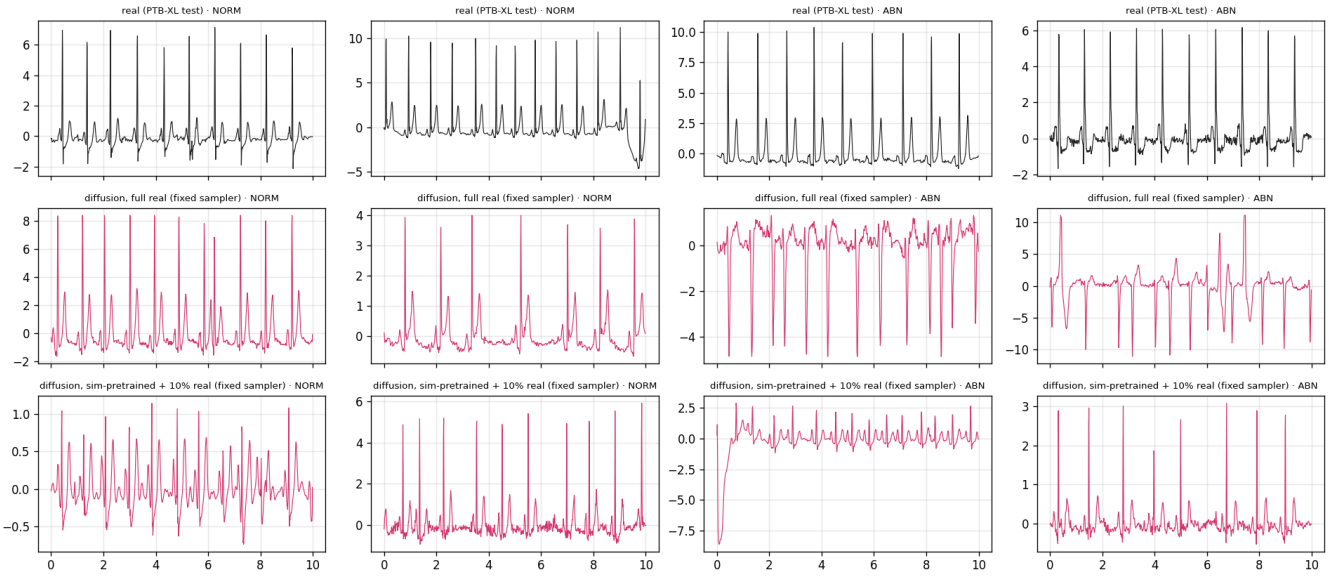


Figure 3: Randomly drawn (not cherry-picked) real test ECG (top) and diffusion samples with the fixed sampler (middle: full data; bottom: simulation-pretrained, 10% real).

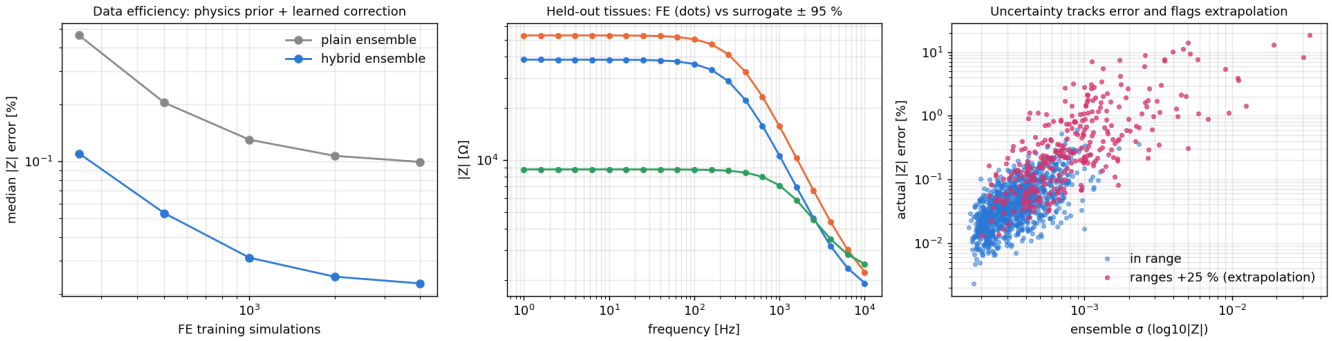


Figure 4: Neural surrogate of the FE electrode solver. Left: data efficiency of the hybrid (physics prior + ensemble) vs. plain ensemble. Middle: held-out spectra, FE (dots) vs. surrogate with 95% bands. Right: ensemble σ vs. actual error, in range and with every parameter range widened by 25%.

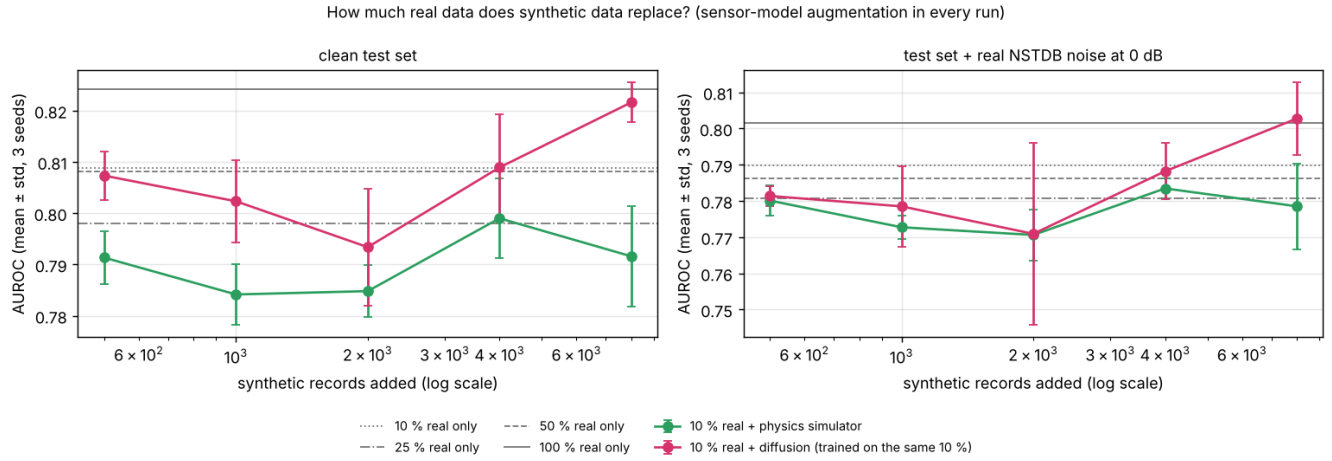


Figure 5: AUROC with 10% real data plus N synthetic records, against real-only training at 10–100% (horizontal lines); 3 seeds.